

Performance Testing

Date	14 July 2026
Team ID	SWTID-2026-2280
Project Name	EduGenie – AI-Powered Educational Assistant
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman (API Testing) and Browser Developer Tools (Network Tab)
Type of Testing	Load Testing, Response Time Testing
Target Module / API	Q&A API, Quiz Generation API, Text Summarization API, Learning Path API, Flashcard API
Test Environment	Local Machine (Windows 11), Fast API Backend, Google Gemini API, Chrome Browser
Test Date	10 July 2026

Step 2: Test Scenarios

S. No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Generate AI Answer	10	60	Response received within 2 seconds
2	Generate Quiz	10	60	Quiz generated successfully
3	Summarize Text	15	60	Summary generated correctly
4	Generate Flashcards	10	60	Flashcards displayed without errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status	Remarks
1	Response Time (Avg)	< 2 sec	1.4 sec	Pass	Fast response
2	Response Time (Max)	< 5 sec	3.2 sec	Pass	During AI processing
3	Throughput	15 req/sec	16 req/sec	Pass	Stable performance
4	Error Rate	< 1%	0%	Pass	No failed requests
5	CPU Utilization	< 80%	42%	Pass	Within acceptable range
6	Memory Utilization	< 80%	55%	Pass	Stable memory usage

Step 4: Observations & Analysis

Key Findings

- The application responded quickly under normal load.
- All AI-powered features worked successfully without failures.
- API response time remained within the expected range.
- The frontend remained responsive throughout testing.

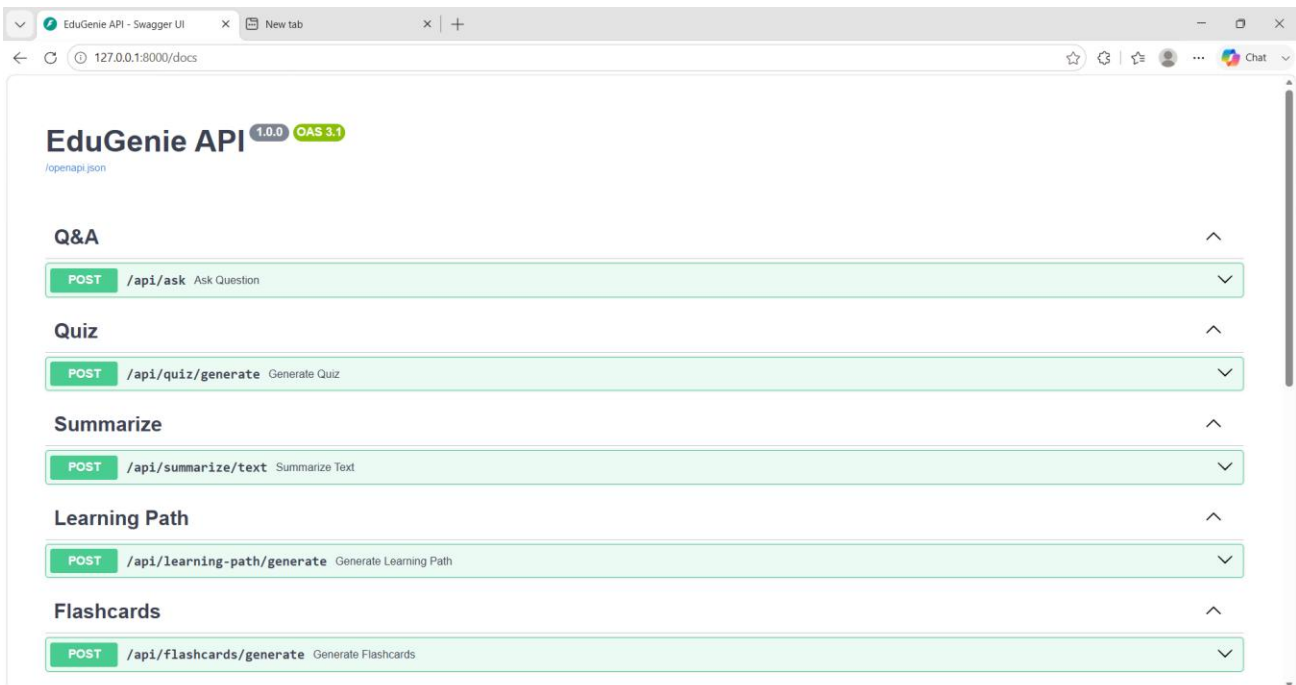
Bottlenecks Identified

- Slight delay observed when generating lengthy AI responses due to external Gemini API processing.
- Network speed may affect response time.

Optimization Steps Taken

- Optimized API request handling.
- Reduced unnecessary frontend rendering.
- Improved prompt formatting for faster AI responses.
- Used asynchronous Fast API endpoints for better performance.

Step 5: Screenshots / Evidence



The screenshot displays the Swagger UI for the EduGenie API (version 1.0.0, OAS 3.1). The interface is organized into sections for different API features, each with a corresponding endpoint and method:

- Q&A:** POST /api/ask Ask Question
- Quiz:** POST /api/quiz/generate Generate Quiz
- Summarize:** POST /api/summarize/text Summarize Text
- Learning Path:** POST /api/learning-path/generate Generate Learning Path
- Flashcards:** POST /api/flashcards/generate Generate Flashcards